

# Data Analysis with Python

## Cheat Sheet: Model Evaluation and Refinement

| Process                                 | Description                                                                                                                                                                                                                                                     | Code Example                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Splitting data for training and testing | The process involves first separating the target attribute from the rest of the data. Treat the target attribute as the output and the rest of the data as input. Now split the input and output datasets into training and testing subsets.                    | <pre> from sklearn.model_selection import train_test_split y_data = df['target_attribute'] x_data=df.drop('target_attribute',axis=1) x_train, x_test, y_train, y_test = train_test_split(x_data, y_data, test_size=0.10, random_state=1) </pre>                                                                                                                                             |
| Cross validation score                  | Without sufficient data, you go for cross validation, which involves creating different subsets of training and testing data multiple times and evaluating performance across all of them using the $R^2$ value.                                                | <pre> from sklearn.model_selection import cross_val_score from sklearn.linear_model import LinearRegression lre=LinearRegression() Rcross = cross_val_score(lre,x_data[['attribute_1']],y_data,cv=n) # n indicates number of times, or folds, for which the cross validation is to be done Mean = Rcross.mean() Std_dev = Rcross.std() </pre>                                               |
| Cross validation prediction             | Use a cross validated model to create prediction of the output.                                                                                                                                                                                                 | <pre> from sklearn.model_selection import cross_val_predict from sklearn.linear_model import LinearRegression lre=LinearRegression() yhat = cross_val_predict(lre,x_data[['attribute_1']], y_data,cv=4) </pre>                                                                                                                                                                              |
| Ridge Regression and Prediction         | To create a better fitting polynomial regression model, like , one that avoids overfitting to the training data, we use the Ridge regression model with a parameter alpha that is used to modify the effect of higher-order parameters on the model prediction. | <pre> from sklearn.linear_model import Ridge pr=PolynomialFeatures(degree=2) x_train_pr=pr.fit_transform(x_train[['attribute_1', 'attribute_2', ...]]) x_test_pr=pr.fit_transform(x_test[['attribute_1', 'attribute_2',...]]) RidgeModel=Ridge(alpha=1) RidgeModel.fit(x_train_pr, y_train) yhat = RidgeModel.predict(x_test_pr) </pre>                                                     |
| Grid Search                             | Use Grid Search to find the correct alpha value for which the Ridge regression model gives the best performance. It further uses cross-validation to create a more refined model.                                                                               | <pre> from sklearn.model_selection import GridSearchCV from sklearn.linear_model import Ridge parameters= [{ 'alpha': [0.001,0.1,1, 10, 100, 1000, 10000, ...]}] RR=Ridge() Grid1 = GridSearchCV(RR, parameters1,cv=4) Grid1.fit(x_data[['attribute_1', 'attribute_2', ...]], y_data) BestRR=Grid1.best_estimator_ BestRR.score(x_test[['attribute_1', 'attribute_2', ...]], y_test) </pre> |



# Skills Network